Tennessee Valley Authority, Post Office Box 2000, Decatur, Alabama 35609-2000

May 29, 2018

10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, D.C. 20555-0001

Browns Ferry Nuclear Plant, Units 1, 2, and 3
Renewed Facility Operating License Nos. DPR-33, DPR-52, and DPR-68
NRC Docket Nos. 50-259, 50-260, and 50-296

Subject: **Licensee Event Report 50-259/2018-003-00**

The enclosed Licensee Event Report provides details of an unanalyzed condition associated with the Browns Ferry Nuclear Plant Emergency Equipment Cooling Water (EECW) System. During postulated fire events where the C3 EECW pump is credited for Control Room Abandonment scenarios, combined with a loss of offsite power, the pump could trip on undervoltage and fail to restart. This would cause a loss of the EECW system's credited function for these events. The Tennessee Valley Authority is submitting this report in accordance with 10 CFR 50.73(a)(2)(ii)(B), as any event or condition that results in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety.

There are no new regulatory commitments contained in this letter. Should you have any questions concerning this submittal, please contact J. L. Paul, Nuclear Site Licensing Manager, at (256) 729-2636.

Respectfully,

(FOR)

D. L. Hughes,
Site Vice President

Enclosure: Licensee Event Report 50-259/2018-003-00 – Unanalyzed Condition due to an Incorrectly Wired Breaker on Emergency Equipment Cooling Water Pump

cc (w/ Enclosure):

NRC Regional Administrator - Region II
NRC Senior Resident Inspector - Browns Ferry Nuclear Plant
NRC Project Manager - Browns Ferry Nuclear Plant

ENCLOSURE

Browns Ferry Nuclear Plant
Units 1, 2, and 3

Licensee Event Report 50-259/2018-003-00

Unanalyzed Condition due to an Incorrectly Wired Breaker on Emergency Equipment Cooling Water Pump

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See Enclosed

CNRC FORM 366 (04-2018)

U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

1. Facility Name	2. Docket Number	3. Page
Browns Ferry Nuclear Plant (BFN), Unit 1	05000259	1 OF 6

4. Title

Unanalyzed Condition due to an Incorrectly Wired Breaker on Emergency Equipment Cooling Water Pump

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved	
Month	Day	Year	Year	Sequential Number	Rev No.	Month	Day	Year	Facility Name	Docket Number
03	29	2018	2018	- 003	- 00	05	29	2018	BFN, Unit 2	05000260
									BFN, Unit 3	05000296

9. Operating Mode: 1

10. Power Level: 100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☒ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER Specify in Abstract below or in NRC Form 366A	

12. Licensee Contact for this LER

Licensee Contact	Telephone Number (Include Area Code)
Baruch Calkin, Licensing Engineer	(256) 614-6713

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to ICES	Cause	System	Component	Manufacturer	Reportable to ICES
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected	15. Expected Submission Date	Month	Day	Year
☐ Yes (If yes, complete 15. Expected Submission Date) ☒ No		N/A	N/A	N/A

Abstract (Limit to 1400 spaces, i.e., approximately 14 single-spaced typewritten lines)

On March 8, 2018, at 0130 Central Standard Time (CST), during a relay functional test of the C3 Emergency Equipment Cooling Water (EECW) Pump trip circuit, incorrect wiring was discovered on the C3 EECW Pump breaker. This condition would have prevented the breaker closing springs from recharging while the transfer switch was in the emergency position. On March 10, 2018, at 2042 CST, Maintenance personnel completed work to correct the wiring for the pump breaker.

On March 29, 2018, at 1344 Central Daylight Time (CDT), BFN Engineering determined that an unanalyzed condition, which significantly affected plant safety, had existed from October 28, 2015 to March 10, 2018. For postulated fire scenarios that could cause a Loss of Offsite Power and also result in Main Control Room Abandonment, the required EECW pumps utilized in the Fire Safe Shutdown procedure may not have been available. On March 29, 2018, at 2128 CDT, Event Notification # 53300 was made to the NRC.

The cause of this condition was a legacy human performance (HU) error involving a failure to properly utilize HU tools. Corrective actions were to correctly rewire the circuitry for the C3 EECW Pump breaker, and to revise plant procedures in order to add HU tools which verify, during testing, that the breaker springs will recharge while the transfer switch is in the emergency position.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 1	05000259	2018	- 003	- 00

NARRATIVE

I. Plant Operating Conditions Before the Event

At the time of discovery, Browns Ferry Nuclear Plant (BFN), Units 1 and 2, were in Mode 1 at approximately 100 percent rated thermal power, and BFN, Unit 3, was in Mode 5 for a refueling outage at zero percent power.

II. Description of Event

A. Event Summary

On March 8, 2018, at 0130 Central Standard Time (CST), during a relay [RLY] functional test of the C3 Emergency Equipment Cooling Water (EECW) [BI] Pump [P] trip circuit, incorrect wiring was discovered on the C3 EECW Pump breaker [BKR]. This condition would have prevented the breaker closing springs from recharging while the transfer switch was in the emergency position.

On March 10, 2018, at 2042 CST, Maintenance personnel completed work to correct the wiring for the pump breaker.

On March 29, 2018, at 1344 CDT, BFN Engineering determined that even though the C3 EECW pump had been unavailable since August 23, 2012, an unanalyzed condition which significantly degraded plant safety actually existed from October 28, 2015 to March 10, 2018. This is based on the fact that the C3 EECW Pump was not credited in the backup control mode until 10CFR50.48(c) – National Fire Protection Association 805 became effective October 28, 2015. For postulated fire scenarios that could cause a Loss of Offsite Power (LOOP) and also result in Main Control Room (MCR) [NA] Abandonment, the required EECW pumps utilized in the Fire Safe Shutdown (FSS) procedure may not have been available. On March 29, 2018, at 2128 CDT this condition was reported to the NRC as Event Notification # 53300.

The Tennessee Valley Authority (TVA) is submitting this report in accordance with 10 CFR 50.73(a)(2)(ii)(B), as any event or condition that results in the nuclear power plant being in an unanalyzed condition that significantly degraded plant safety.

B. Status of structures, components, or systems that were inoperable at the start of the event and that contributed to the event

There were no structures, components, or systems (SSCs) that were inoperable at the time of discovery, which contributed to this condition.

NARRATIVE

C. **Dates and approximate times of occurrences**

Date	Time	Event
August 23, 2012	1247 CDT	Work complete on Work Order 113567386. This was the last work performed on the C3 EECW Pump breaker prior to discovery of this condition.
March 8, 2018	0130 CST	Incorrect wiring discovered on C3 EECW Pump breaker during a relay functional test of the trip circuit.
March 10, 2018	2042 CDT	Maintenance personnel completed work to correct the field wiring for the pump breaker.
March 29, 2018	1344 CDT	A Past Operability Evaluation (POE) determined that an unanalyzed condition existed which affected the C3 EECW Pump.
March 29, 2018	2128 CDT	Event Notification # 53300 made to the NRC.

D. **Manufacturer and model number of each component that failed during the event**
This condition did not involve any failed components.

E. **Other systems or secondary functions affected**
There were no other systems or secondary functions affected by this condition.

F. **Method of discovery of each component or system failure or procedural error**
There were no component failures, system failures, or procedural errors associated with this condition.

G. **Failure mode, mechanism, and effect of each failed component**
This condition did not involve any failed components.

H. **Operator actions**
There were no operator actions taken as a result of this condition.

I. **Automatically and manually initiated safety system responses**
There were no safety system responses initiated as a result of this condition.

NARRATIVE

III. Cause of the Event

A. Cause of each component or system failure or personnel error

Human performance tools which could have prevented this condition were not utilized.

B. Cause(s) and circumstances for each human performance related root cause

This issue was a legacy human performance issue which could have been prevented through the use of the "Stop When Unsure" HU tool. The use of the "Stop When Unsure" HU tool has since been covered under the Procedure Adherence rollout in 2017.

IV. Analysis of the Event

The safety objective of the EECW System is to provide cooling water to the Standby Diesel [DG] engine [ENG] coolers [LB], the Residual Heat Removal [BO] pump seal coolers and pump room coolers, the Core Spray [BG] pump room coolers, the Unit 3 Control Bay chillers [CHU], Unit 3 Shutdown Board (SDBD) [ECBD] Room Chillers, Unit 3 Electrical Board Room Air Conditioning Units, and additional makeup for the fuel pool. There are four service water pumps dedicated to the EECW system that supply two independent headers.

The C3 EECW Pump Breaker was wired in an incorrect manner that prevented the breaker closing springs from recharging while the transfer switch was in the emergency position. The manual action of taking the transfer switch to the emergency position would occur locally at the breaker if the MCR was abandoned during an event resulting in an entry into the Fire Safe Shutdown (FSS) procedures. Because of the incorrect wiring, no wiring connection existed from the emergency DC control power to the closing spring charging motor. As a result, the breaker closing springs would be unable to be recharged with the transfer switch in the emergency position. Since power was available to the close charging spring with the transfer switch in the normal position, the closing spring would have been charged when the operators placed the transfer switch in emergency. With the spring charged when the transfer switch was taken to emergency, the breaker could be closed one time. However, there are postulated scenarios where the breaker would have to close more than once with the transfer switch in emergency. The closing spring would not have recharged resulting in unavailability of the C3 EECW pump.

BFN FSS procedures for MCR Abandonment require that at least two EECW pumps are in service, and include steps to align the A3 and C3 EECW Pumps from Backup Control. Assuming the C3 EECW pump had to be started once with the transfer switch in emergency, a subsequent LOOP during a MCR Abandonment scenario could have resulted in a condition where the C3 EECW Pump would trip and fail to restart, causing a loss of safety function for a required EECW pump during a fire event.

NARRATIVE

V. Assessment of Safety Consequences

A Probabilistic Risk Assessment performed by TVA determined that the risk significance of this condition was negligible, corresponding to a maximum change in Core Damage Frequency of less than 1E-6/yr and change in Large Early Release Frequency less than 1E-7/yr. Therefore, there is no significant impact to the health and safety of the public from this condition.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event

Two EECW pumps are required for aligning the EECW system from backup control. The FSS procedure for Units 1, 2, and 3 includes steps for aligning A3 and C3 EECW pumps from backup control. While 4160V Shutdown Board C would be aligned for abandonment and the B3 EECW Pump would have been available, the FSS procedure for control room abandonment does not contain steps for starting the B3 EECW Pump from backup control.

The POE determined that even with the C3 EECW backup control function lost, the system was not operated outside the TS limitations since the B3 and D3 EECW Pumps and their backup control functions would have been restored within seven days per TS LCO 3.7.2.

B. For events that occurred when the reactor was shut down, availability of systems or components needed to shut down the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident

During the period when this condition existed there were multiple time periods when one or more BFN reactors were shut down. However, the EECW system's Backup Control functions are only required to be Operable in Modes 1 and 2.

C. For failure that rendered a train of a safety system inoperable, an estimate of the elapsed time from the discovery of the failure until the train was returned to service

This condition did not result in the TS inoperability of any safety system. However, this condition rendered the C3 EECW Pump, which is credited for some fire scenarios, unavailable for the purposes of those scenarios.

VI. Corrective Actions

This condition was entered into the TVA Corrective Action Program (CAP) and is being tracked under Condition Report 1394604.

1. FACILITY NAME	2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Browns Ferry Nuclear Plant, Unit 1	05000259	2018	- 003	- 00

NARRATIVE

A. Immediate Corrective Actions

The immediate corrective action was to correctly rewire the circuitry for the C3 EECW Pump breaker. The circuitry was repaired on March 10, 2018.

B. Corrective Actions to Prevent Recurrence or to reduce probability of similar events occurring in the future

Plant procedures will be revised in order to add HU tools which verify, during testing, that the breaker springs will recharge while the transfer switch is in the emergency position.

VII. Previous Similar Events at the Same Site

A review of the BFN CAP and Licensee Event Reports for Units 1, 2, and 3 found no instances within the past five years of degraded or unanalyzed conditions on the EECW Pumps which resulted from improper wiring.

VIII. Additional Information

There is no additional information.

IX. Commitments

There are no new commitments.